THOMSON COURSE TECHNOLOGY

Guide to Network Defense and Countermeasures Third Edition

Chapter 6

Wireless Network Fundamentals

Objectives

- Explain wireless communication concepts
- Describe radio frequency characteristics relevant to wireless networking

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Wireless Communications Primer

- **Wireless networking**: any exchange of data between computers and other devices that does not use cables
- Different from cabled networks:
  - Use certain types of electromagnetic radiation
    - Radio frequency (RF) waves is most commonly used
    - Infrared (IR) radiation used mainly for communication with peripheral devices

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Electromagnetic Radiation, Part 1

- **Electromagnetic (EM) radiation**: electromagnetic energy traveling as a self-propagating wave and spreading out at the same time
- **Wave**: means of transporting energy from one place to another
  - Energy is transported by a disturbance that occurs in a distinct repeating pattern
- **Amplitude**: maximum departure of a wave from the undisturbed state
- **Frequency**: number of times an event occurs in a specified period (measured in hertz)

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Electromagnetic Radiation, Part 2

- **Wavelength**: distance between repeating units of the wave (usually the midpoint or crest)
- Frequency has an inverse relationship with wavelength
  - Frequency is number of waves per second
  - Wavelength is the distance between waves

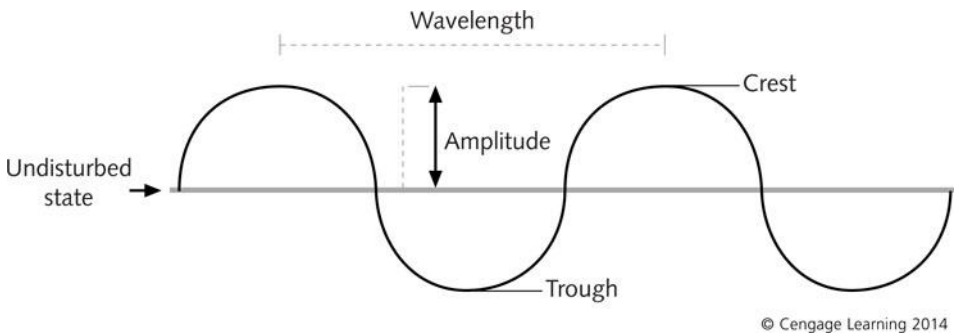

Figure 6-1 Wave properties

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Infrared Transmissions, Part 1

- **Infrared transmissions** use infrared light pulses
  - Require an emitter (laser diode or LED) and a detector (sometimes combined with an emitter)
  - Intensity of the light pulse indicates the on or off status of each bit of data
- **Directed IR transmission**: requires emitter and detector to be pointed directly at one another
- **Diffused IR transmission**: relies on reflected light that can bounce off walls or other objects

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Infrared Transmissions, Part 2

- Advantages of IR wireless:
  - Does not interfere with other signals and is not susceptible to interference from them
  - IR signals cannot pass through walls
- Disadvantages of IR wireless:
  - Limited range
  - Low speeds of up to 4 Mbps
  - Requires direct line of sight or in-the-room conditions

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Radio Frequency Transmissions

- RF is the most commonly used transmission medium for WLANs
- RF can travel through walls and travel great distances
- RF involves transmission ranges, signal modulation, and interference
  - More complex than IR

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Common RF Bands

Band and abbreviation	Frequency	Common uses
Extremely low frequency (ELF; maritime use)	3–30 Hz	Naval and maritime
Super-low frequency (SLF; maritime use)	30–300 Hz	Naval and maritime
Ultra-low frequency (ULF)	300–3000 Hz	Communication in mines
Very low frequency (VLF)	3–30 KHz	Communication with near-surface
submarines		
Low frequency (LF)	30–300 KHz	Cordless telephones
Medium frequency (MF)	300 KHz–3 MHz	AM radio
High frequency (HF)	3–30 MHz	CB and shortwave radio
Very high frequency (VHF)	30–144 MHz 144–174 MHz 174–328.6 MHz	FM radio TV channels 2–6 Taxi radios TV channels 7–13
Ultra-high frequency (UHF)	328.6–806 MHz 806–960 MHz 960 MHz–2.3 GHz 2.3–2.9 GHz	Public safety Cell phones Air traffic control radar WLANs
Super-high frequency (SHF)	2.9–30 GHz	WLANs
Extremely high frequency (EHF)	30–300 GHz	Radio astronomy
Atmosphere blocks frequencies	Above 300 GHz	N/A

Table 6-1 Common RF bands

Transmission Ranges

- Transmission ranges vary depending on the standard in use and environment
- Generally, lowering bandwidth increases coverage area
  - The rate at which a wireless client receives data decreases as client moves away from transmitter
- **Access point**: an electronic device that connects to a wired network and can transmit and receive wireless signals
  - Enforcing security for wireless signals requires careful placement of APs

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Interference

- **Co-channel interference** occurs when signals from APs interfere with each other
  - Must arrange APs so that overlapping signals do not share the same channel (frequency)
- Interference
  - RF can be highly susceptible to interference from electrical storms, solar activity, laser printers, and other forms of EM radiation (microwave ovens)
  - **Multipath**: a signal has more than one path from transmitter to receiver
    - If signal is reflected, the reflected path can interfere with direct path (this problem is called **fading**)

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Radio Frequency Signal Behavior, Part 1

- RF signal behavior is characterized by whether a factor contributes to an increase (gain) or decrease (loss) in power
  - Gain: positive difference in amplitude between two signals
    - Achieved by magnifying the signal
  - Loss: negative difference in amplitude of signals (sometimes called attenuation)
- Common factors that result in loss:
  - Absorption – when certain types of material absorb RF signals, such as wood, concrete, and asphalt
  - Reflection – when RF signals bounce off some materials

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Radio Frequency Signal Behavior, Part 2

- Common factors that result in loss (continued):
  - Scattering – when small objects and rough textures disperse signals
  - Refraction – when differences in density between air masses over distances cause problems (signals may bend instead of traveling in a straight line)
  - Diffraction – similar to refraction, except signal bends around an object in its path
  - Voltage standing wave ratio (VSWR) – caused by differences in equipment rather than external influences

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Measuring RF Signals, Part 1

- RF power is measured on a linear scale using milliwatts (mW)
  - Watt: measure of power or the rate at which work is done
  - One mW is equal to one-thousandth of one watt
- Decibel-milliwatts (dBm) is the reference point that relates the decibel scale to the linear milliwatt scale
  - Specifies that 1 mW = 0 dBm
  - RF power gains and losses on a relative scale are measured in decibels (dB) instead of mW

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The 10s and 3s Rules of RF Math

–3 dB equals	half the power in mW
+3 dB equals	double the power in mW
–10 dB equals	one-tenth the power in mW
+10 dB equals	10 times the power in mW

Table 6-3 The 10s and 3s rules of RF math

Measuring RF Signals, part 2

- Equivalent Isotropically Radiated Power (EIRP): power radiated by a wireless system's antenna
  - Uses a measurement known as isotropic decibels (dBi) that applies only to an antenna's gain
- Transmitter Power Output (TPO) measures the power being delivered to the transmitting antenna

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RF Signaling

- RF transmits a carrier signal
  - Changes based on the signal's voltage and direction
- RF data is transmitted as analog or digital signals
  - Analog RF signal: continuous wave that oscillates between positive and negative voltage
    - Must be converted into digital format
  - Digital RF signal: divided into discrete segments or defined states within the carrier's range
- **Modulation**: changing characteristics of the signal
- Three characteristics of a carrier signal can be modified to enable it to carry data: height, frequency, and relative starting point of the signal

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Analog Modulation

- Analog modulation methods:
  - Amplitude modulation (AM) – the height of the carrier wave is changed so a higher wave represents a 1 bit and a lower wave represents a 0 bit
  - Frequency modulation (FM) – number of waves representing one cycle is changed so that the number representing a 1 bit is greater
  - Phase modulation (PM) – cycle's starting point is changed when the bit being transmitted changes from 1 to 0

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Analog Modulation Techniques

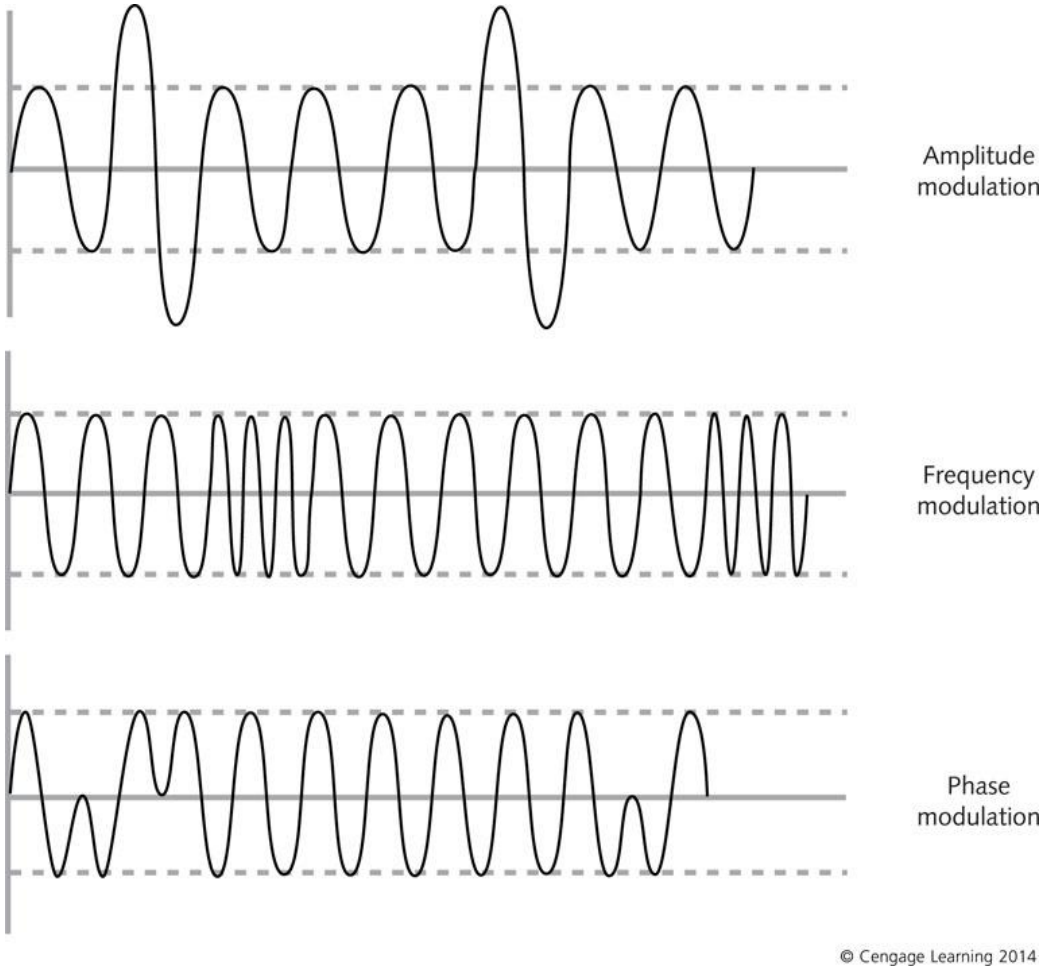

Figure 6-3 Analog modulation techniques

Digital Modulation, Part 1

- Digital modulation techniques are superior to analog methods for four reasons:
  - More efficient use of bandwidth
  - Fewer interference problems
  - Error correction that is more compatible with other digital systems
  - Less power required to transmit

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Digital Modulation, Part 2

- Three binary signaling techniques:
  - *Return-to-zero (RTZ)* – Voltage increases to represent a 1 bit, no voltage represents a 0 bit
    - Voltage for a 1 bit drops back to zero before the end of the bit period
  - *Non-return-to-zero (NRZ)* - Voltage increases to represent a 1 bit, no voltage represents a 0 bit
    - Voltage for a 1 bit does not drop back to zero before the end of the bit period
  - *Polar non-return-to-zero (polar NRZ)* – Voltage increases to represent a 1 and drops to negative voltage to represent a 0 bit

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Digital Modulation, Part 3

- RF signals are narrowband transmissions
  - Transmit on one frequency or small frequency range
- Common digital modulation methods:
  - Amplitude shift keying (ASK) – height of the carrier can be changed to represent a 1 or 0 bit
  - Frequency shift keying (FSK) – carrier signal's frequency is changed to represent a 1 or 0 bit
  - Phase shift keying (PSK) – similar to phase modulation
  - Frequency division multiplexing (FDM) – multiple base signals are modulated on different carrier waves and combined to form a composite signal

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Narrowband Transmission

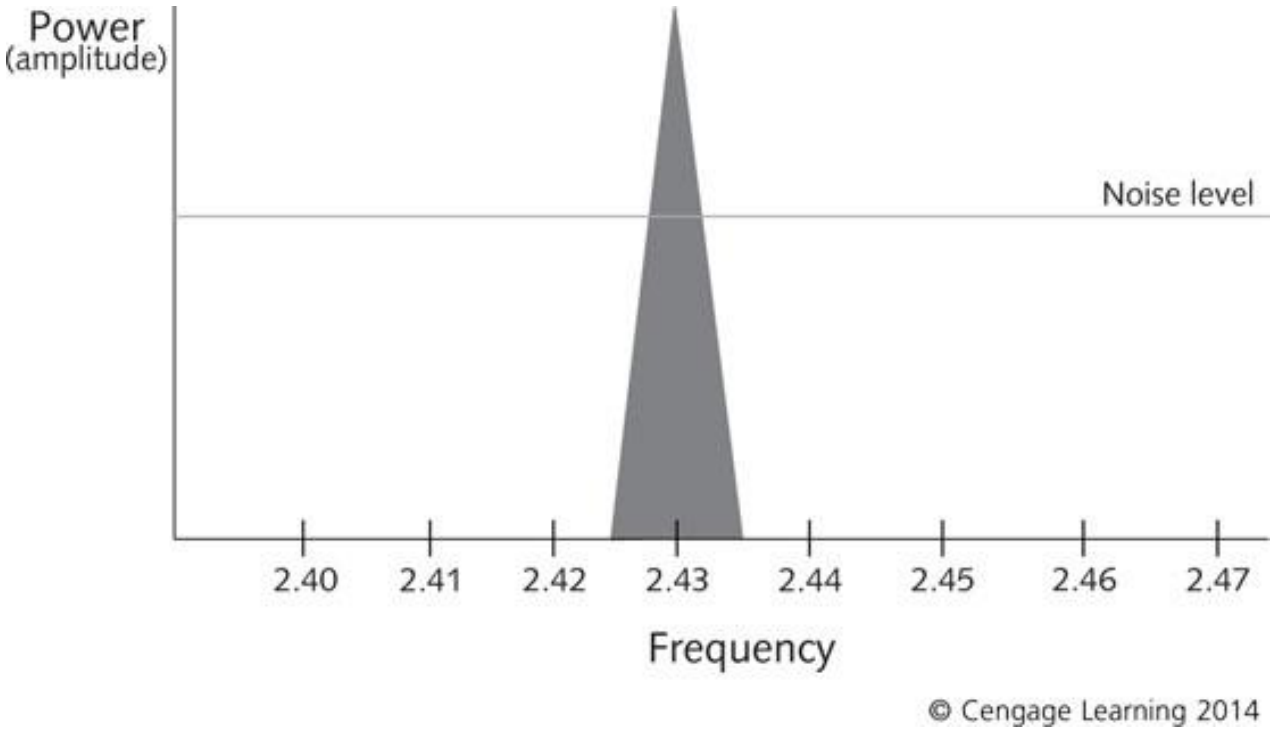

Figure 6-4 Narrowband transmission

Spread Spectrum, Part 1

- Spread spectrum spreads a signal over a broader portion of the radio band
- Advantages of spread spectrum over narrowband:
  - Bandwidth of signal is higher than original message
  - Bandwidth is determined by the spreading function
    - Known only to the transmitter and receiver
- In spread spectrum:
  - The spreading function attaches a key (called a spreading code or sequence) to the communication channel

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Spread-spectrum Transmission

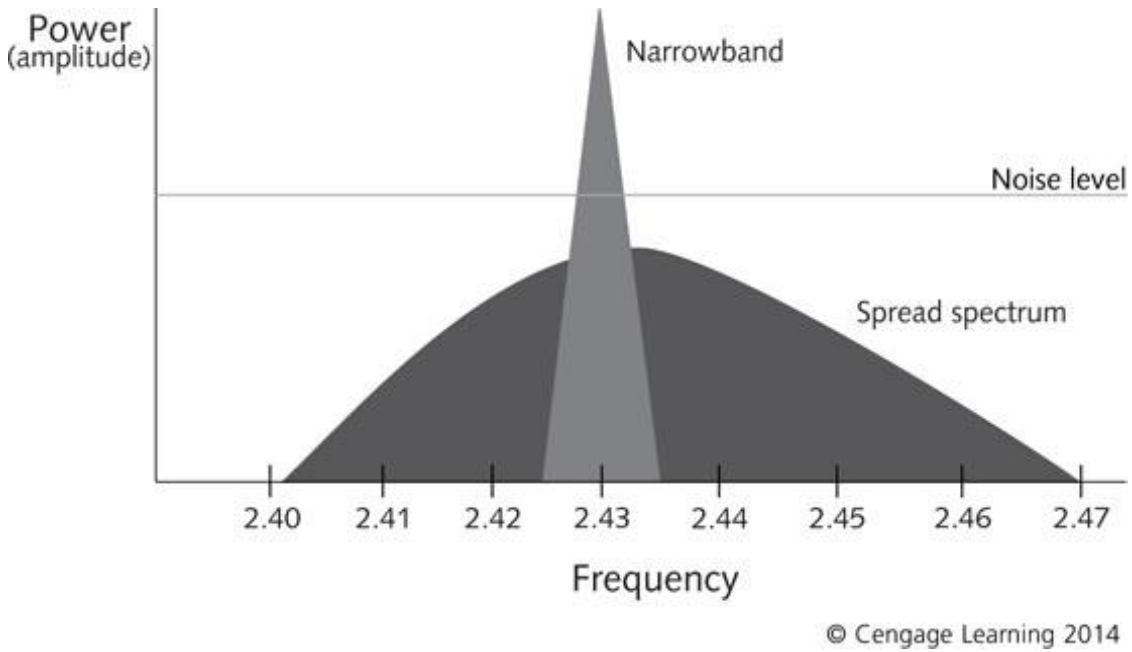

Figure 6-5 Spread-spectrum transmission

Spread Spectrum, Part 2

- Major methods of spread spectrum:
  - **Direct sequence spread spectrum (DSSS)** – key is applied at the data level
  - **Frequency hopping spread spectrum (FHSS)** – key is applied at the carrier frequency level
  - **Orthogonal frequency division multiplexing (OFDM)** – high-speed signal is divided into smaller pieces and sent simultaneously across lower-speed channels

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DSSS Transmission

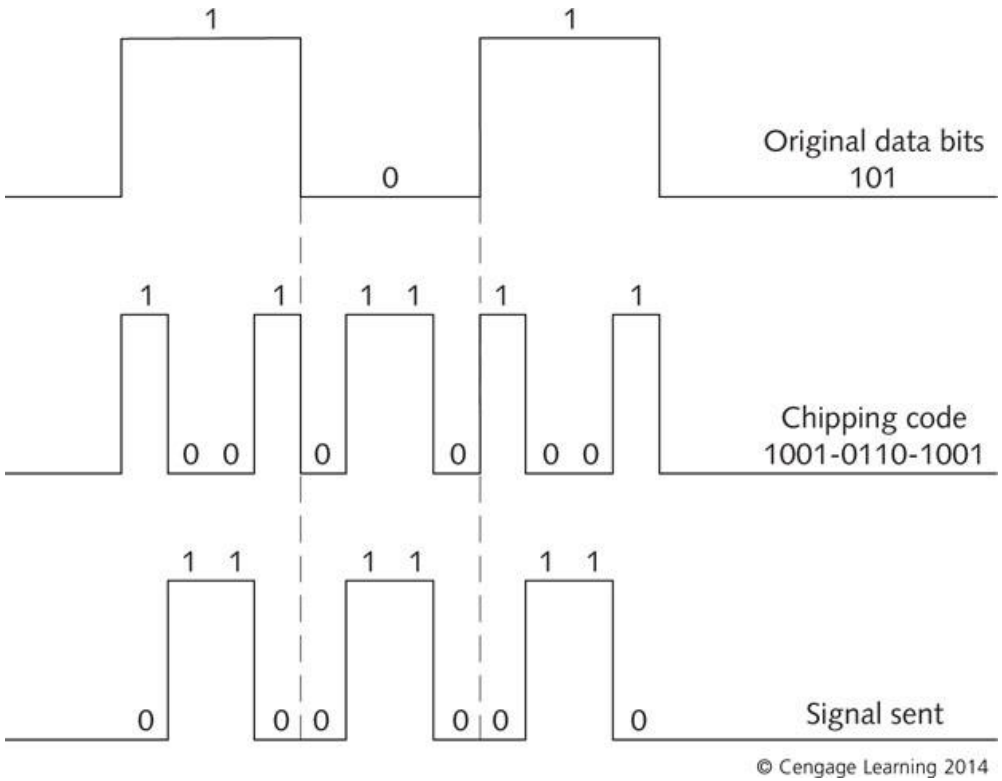

Figure 6-6 DSSS transmission

FHSS Transmission

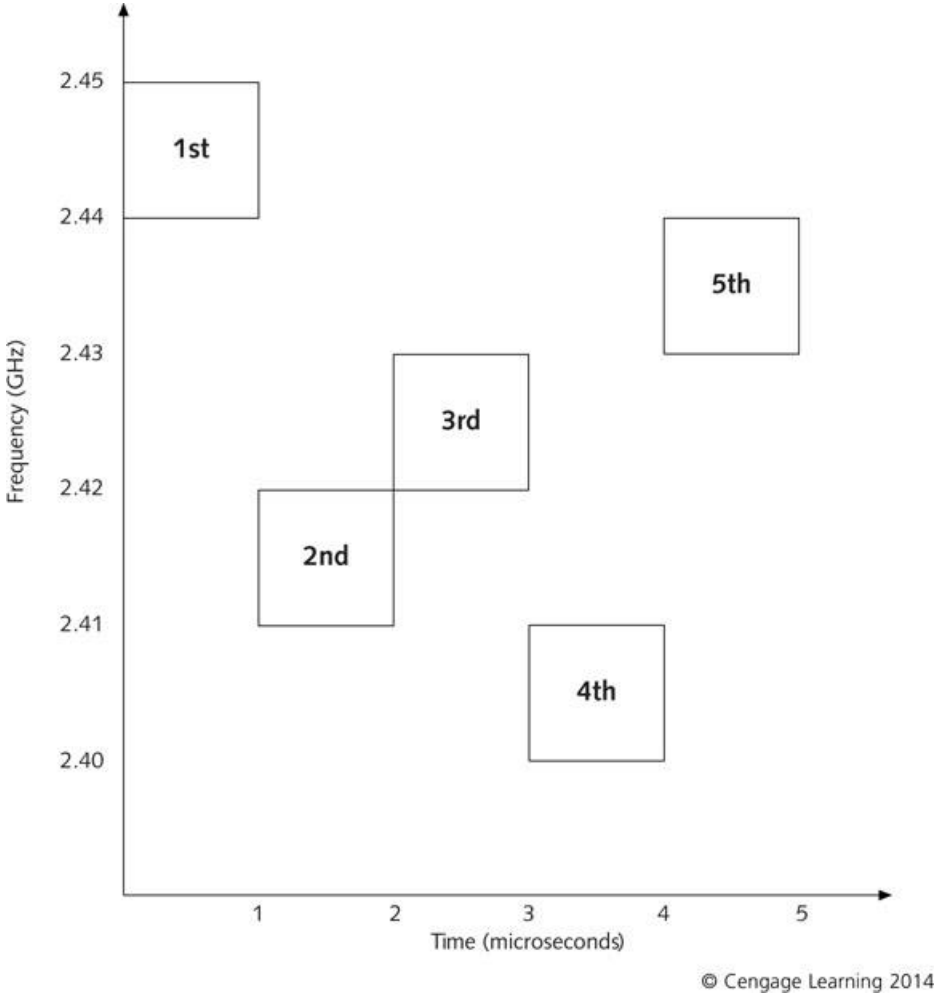

Figure 6-7 FHSS transmission

Spread Spectrum, Part 3

- In DSSS, an expanded redundant chipping code is used to transmit each bit
  - Chipping code: term for bit pattern
  - DSSS is less vulnerable to data loss from interference but requires high bandwidth
- In FHSS, carrier hops frequencies over a wide band according to a sequence defined by the key
  - Key is called the hopping code and it determines the sequence and speed of frequency hops
  - Advantages of FHSS are immunity to jamming and interference and it is secure

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Wireless LANs and Their Components

- To secure a WLAN, you need to be familiar with:
  - Wireless components
  - Topologies
  - Transmission and frequency ranges
  - Methods of identifying and eliminating interference sources

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Wireless NICs

- When a wireless NIC (WNIC) prepares to transmit, it does the following:
  - Changes the computer's internal data from parallel to serial transmission
  - Divides data into packets and attaches address information
  - Determines where to send the packet
  - Transmits the packet

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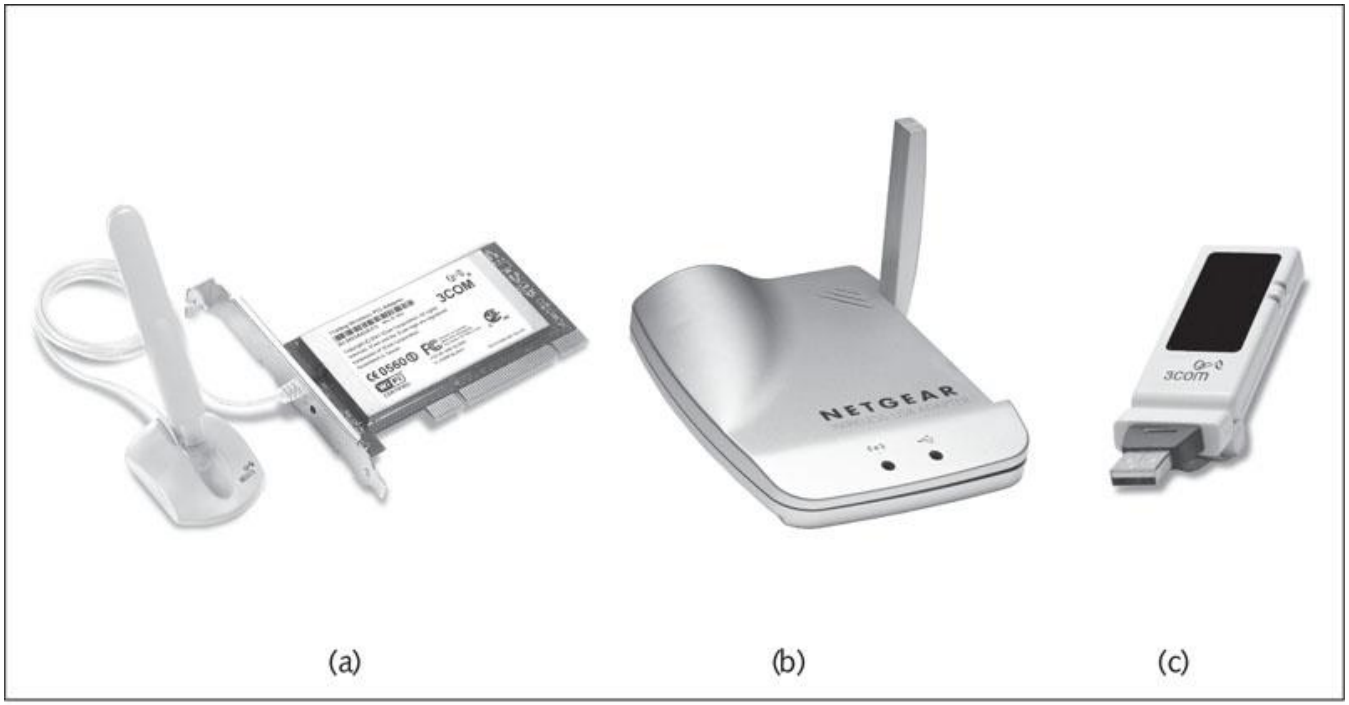
Desktop Computer WNICs



Sources: Hewlett-Packard, Netgear

Figure 6-8 Desktop computer WNICs

Access Points

- Access point (AP) - an antenna and radio transceiver used to transmit and receive signals and to perform the following functions:
  - Acts as a base station for the wireless network segment
  - Serves as the bridge between wired and wireless segments
- Preferred placement of APs is on the ceiling or high on a wall
  - Solution to getting power to APs placed in ceilings or up high: **Power over Ethernet (PoE)**
    - PoE: power for AP unit is supplied through unused wires in Ethernet cabling

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Wireless Access Point

Source: Cisco

Figure 6-9 Wireless Access Point

Antennas, Part 1

- RF waves are transmitted and received by an antenna
- EIRP is the measurement of total power radiated by a wireless system's antenna
  - FCC uses the term intentional radiator to describe a device designed to generate radio signals
- Fundamental characteristics of antennas:
  - As frequency gets higher, wavelength gets smaller (requiring a smaller antenna)
    - Antenna length should be ¼ of the wavelength
  - As antenna gain increases, coverage area narrows

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Antenna Sending and Receiving Radio Signals

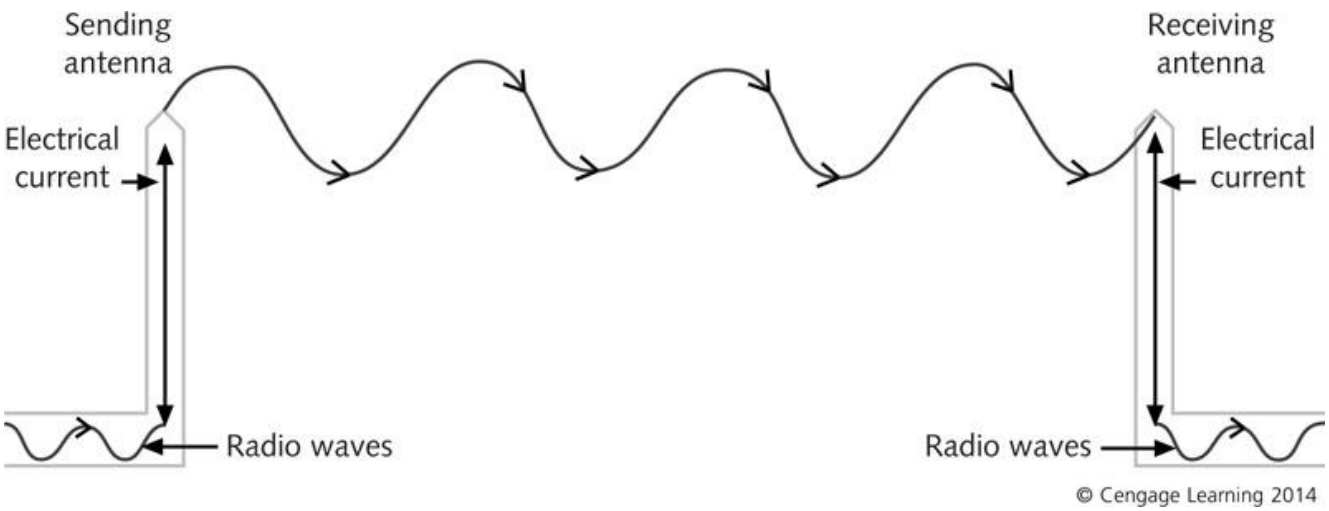

Figure 6-10 Antenna sending and receiving radio signals

Antennas, Part 2

- Other characteristics of RF antenna transmissions:
  - Polarization – plane in which radio waves propagate or the orientation of radio waves as they leave the antenna
  - Wave propagation – dispersal pattern of waves as they travel from sending to receiving antennas
  - Fresnel zone – series of ellipsoidal shapes in the wave calculated to determine the signal strength
    - Also identifies potential obstacles and multipath distortion between antennas
  - Free space path loss – phenomenon of signals dispersing as they travel from the sending antenna
    - Signal becomes weaker

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The Fresnel Zone

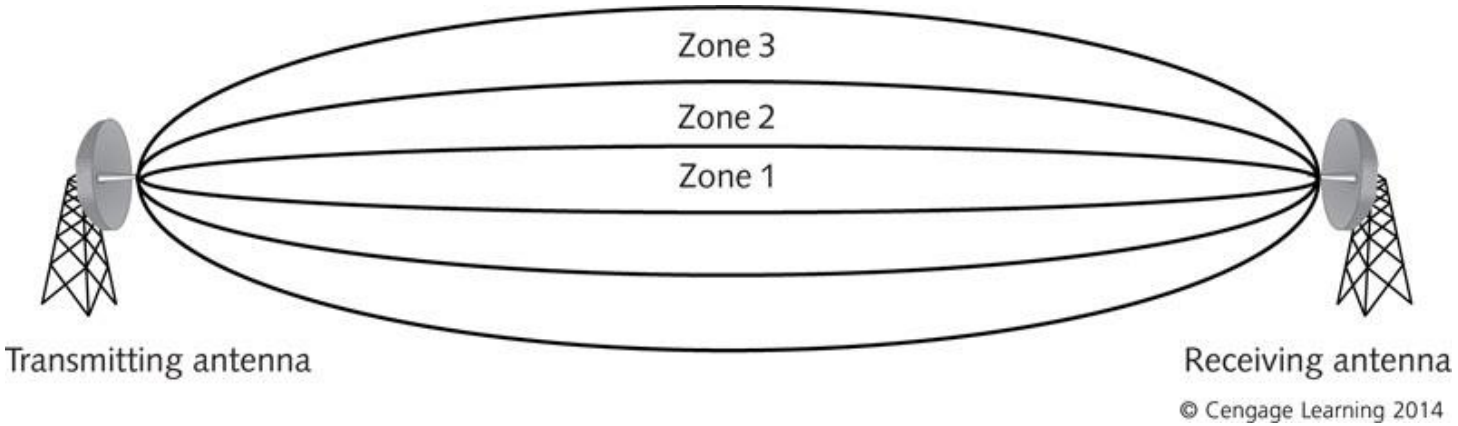

Figure 6-11 The Fresnel zone

Basic Antenna Types

- There are three basic types of antennas: omnidirectional (also known as dipole), semidirectional, and highly directional

Type of antenna	Radiation direction	WLAN installation
Omnidirectional	All directions equally	Most common for a WLAN AP in one building
Semidirectional	Focuses energy in one direction	Used for short- or medium-range remote wireless bridges, such as a point-to-point link between neighboring buildings
Highly directional	Tightly focuses signal beam in one direction	Long-distance, point-to-point links, as in a wireless link between buildings separated by up to 25 miles; not commonly used for WLANs

Table 6-4 Basic antenna types

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Remote Wireless Bridges

- Remote wireless bridges connect wired and wireless segments like APs, with two exceptions:
  - Transmits at higher power than an AP
  - Uses a directional antenna to focus transmission in one direction
    - APs use omnidirectional transmission
- Operates in four modes:
  - Access point mode
  - Root mode
  - Nonroot mode
  - Repeater mode

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Point-to-Point Wireless Bridging

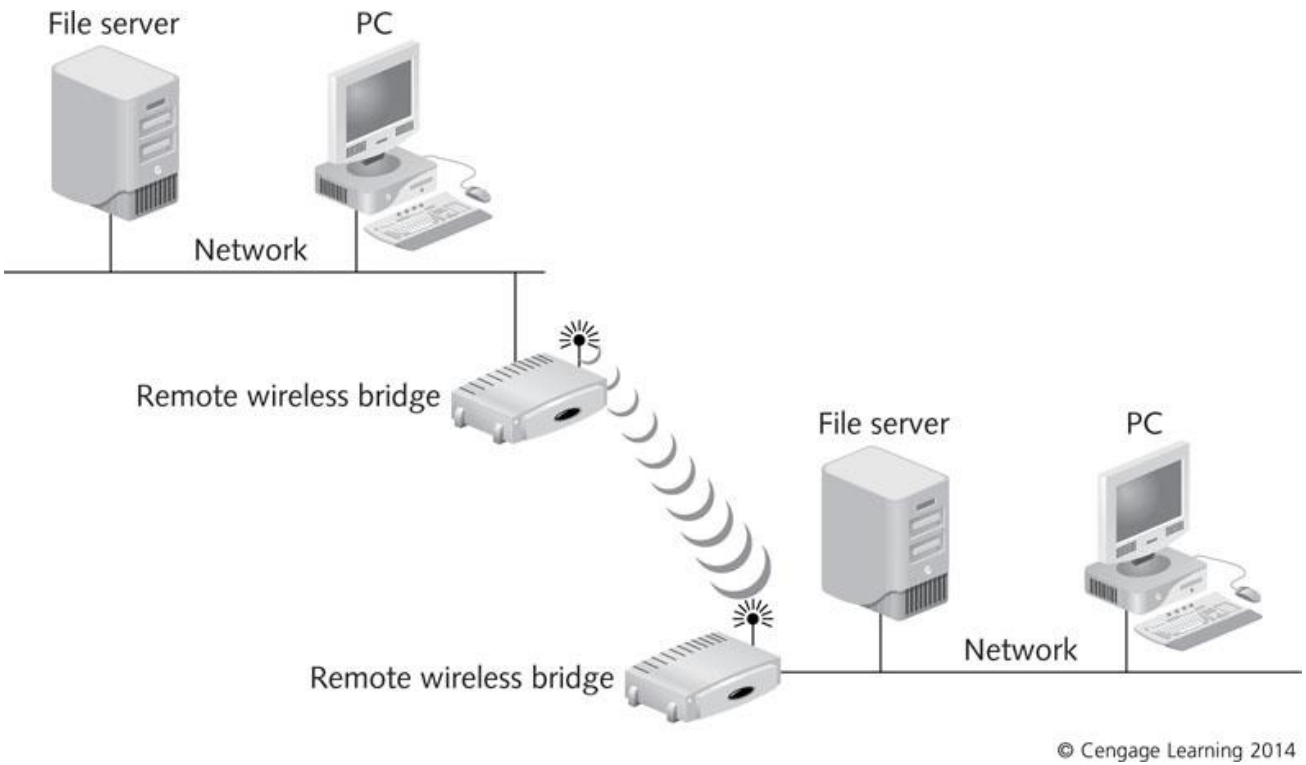

Figure 6-12 Point-to-point wireless bridging

Point-to-Multipoint Wireless Bridging

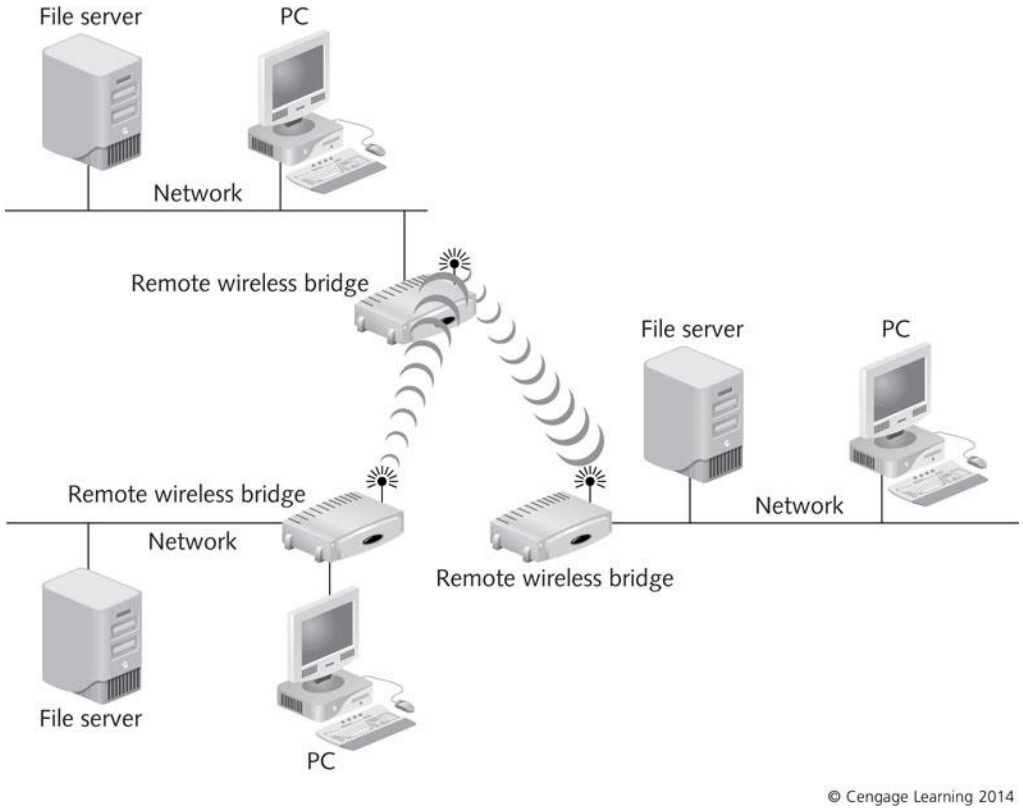

Figure 6-13 Point-to-multipoint wireless bridging

Wireless Gateways

- Wireless gateway combines management and security into a single appliance
- Can perform the following functions:
  - Authentication
  - Encryption
  - Intrusion detection
  - Malicious program protection
  - Bandwidth management
  - Centralized network management

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WLAN Configurations

- Three basic WLAN configurations:
  - Basic Service Set (BSS) – group of wireless devices are served by a single AP
    - Must be assigned a unique identifier known as the service set identifier (SSID)
    - Geographical coverage is called the Basic Service Area (BSA)
  - Extended Service Set (ESS) – APs are set up to provide overlap
    - Coverage areas are called cells and movement between cells is called roaming
  - Independent Basic Service Set (IBSS) – wireless network that does not use an AP

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ESS Configuration

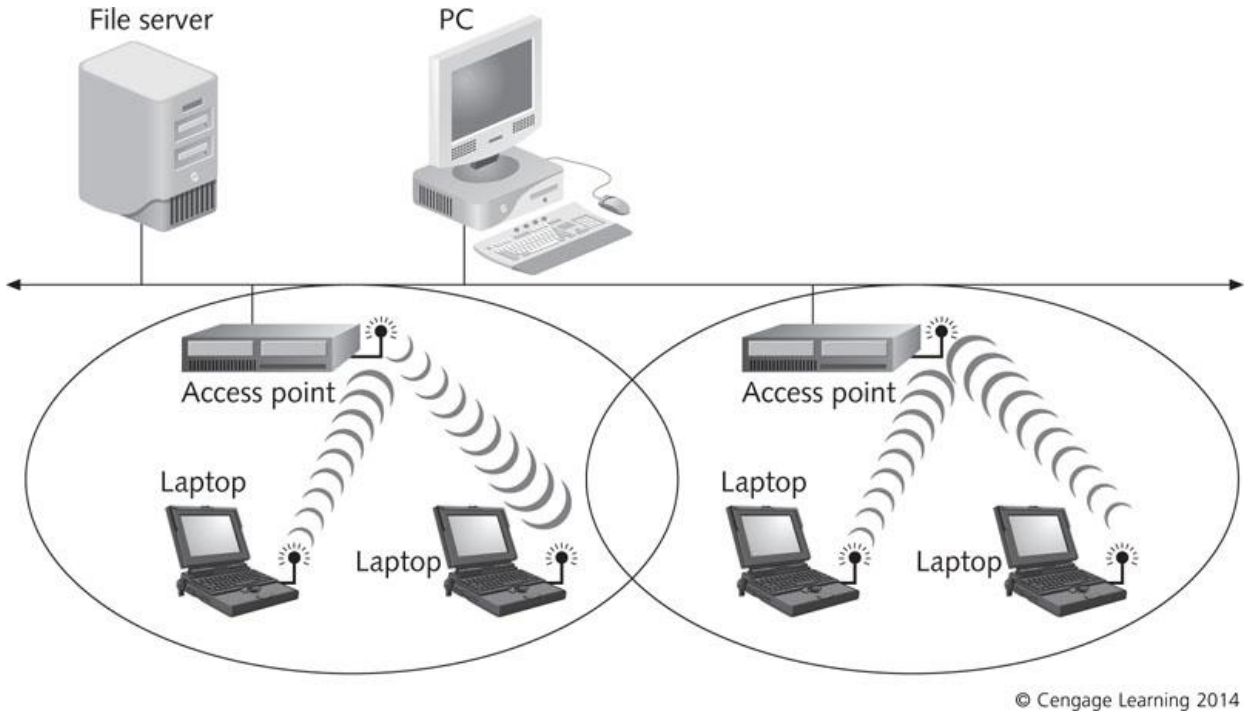

Figure 6-15 ESS configuration

Wireless Networking Standards

- Wireless networking technology was developed in a haphazard way
  - Different companies worked on similar problems and came up with different solutions
- Wireless standards process has become more efficient
  - Still overlaps and uncertainty as wireless networking expands

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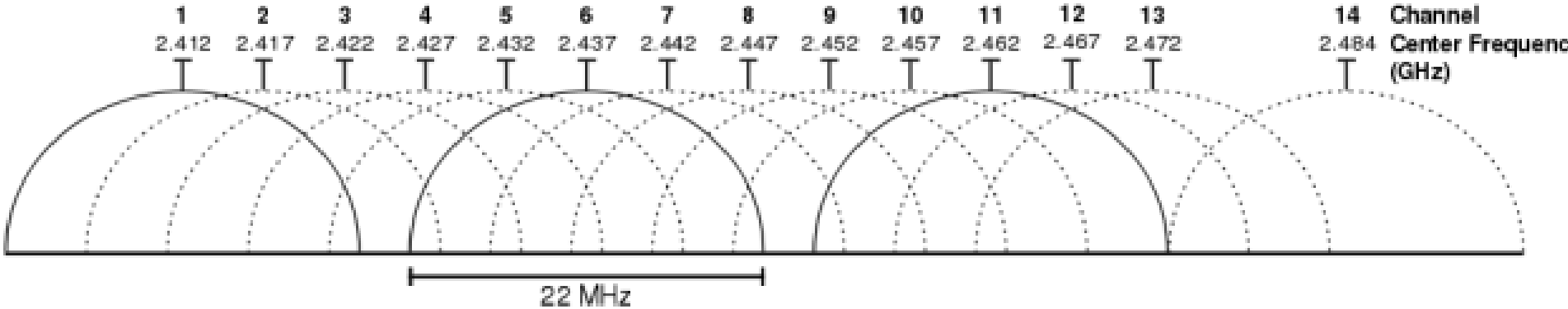
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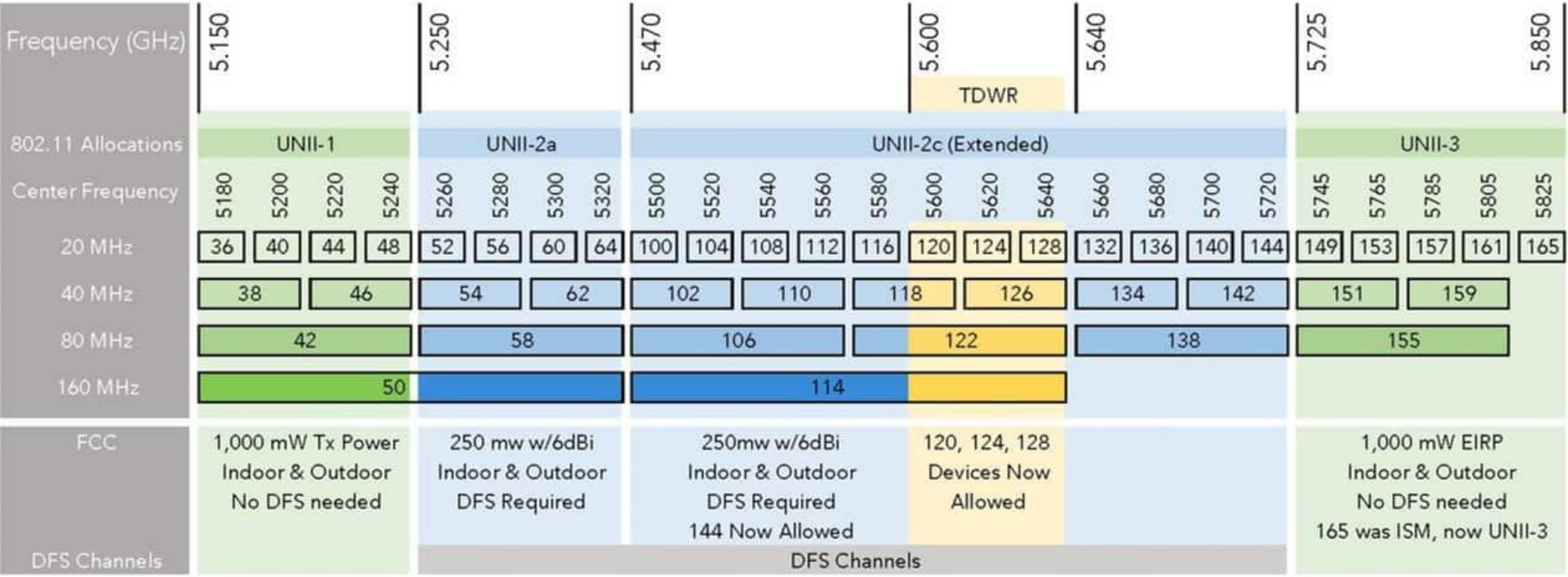
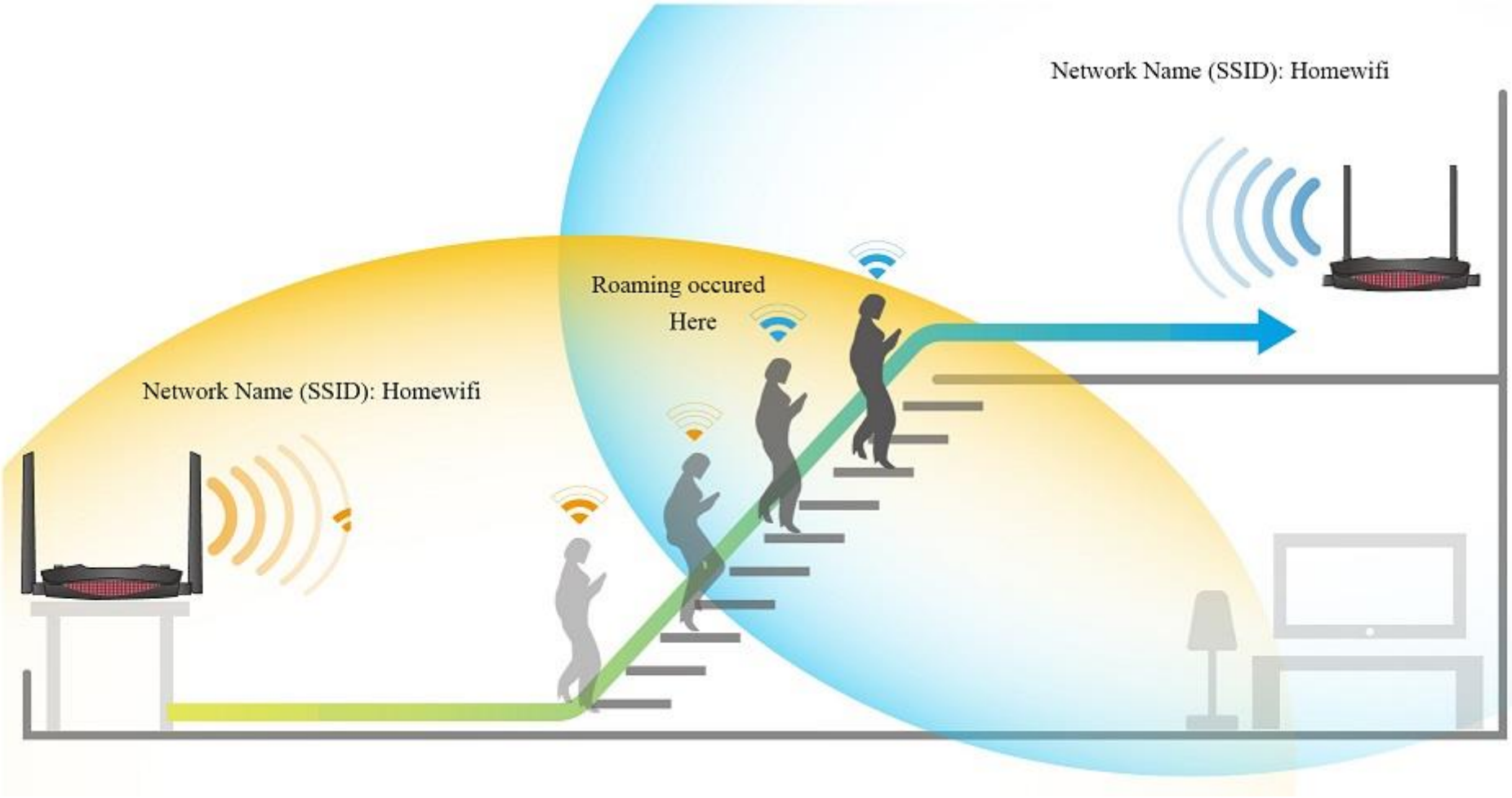
22 MHz

5 GHz Channel Allocations




Network Name (SSID): Homewifi
Roaming occured
Here
Network Name (SSID): Homewifi

IEEE 802.11, Part 1

- IEEE 802.11 – first released in 1997
  - Most recent iteration is IEEE Std. 802.11-2007
    - Includes all ongoing amendments up to that time
    - Since 2007, 802.11n (2009) have been added
- IEEE 802.11b (1999) – ratified before 802.11a
  - Operates in the 2.4 GHz band and maximum bandwidth supported is 11 Mbps
  - No longer used in contemporary WLANs
- IEEE 802.11a (1999) – ratified after 802.11b
  - Operates in the 5 GHz band
    - Not subject to interference by microwave ovens and cordless phones that operate in 2.4 GHz range
    - Maximum bandwidth is 54 Mbps

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IEEE 802.11, Part 2

- 802.11g (2003) – operates in the 2.4 GHz band
  - Interoperable with 802.11a devices
  - Maximum bandwidth is 54 Mbps
- 802.11i (2004) – wireless security standard
  - WPA 2 was released to map exactly to the 802.11 standard
- 802.11r (2008) – designed to provide **fast basic service set transition (FT)**
  - Involves having a client perform a security association with the next AP before the client leaves the range of the current AP

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IEEE 802.11, Part 3

- 802.11n (2009) – defines a standard that supports multiple-input multiple-output (MIMO)
  - Uses both 2.4 GHz and 5 GHz radio frequencies to simultaneously send or receive data
  - Maximum bandwidth can reach 450 Mbps
- 802.11v (2011) – defines standards that allow wireless stations to exchange operational information to improve wireless network performance

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- 802.11ac– will use the 5 GHz band
  - Current home wireless routers are likely 802.1ac-compliant, and operate in the 5 GHz frequency space. With Multiple Input, Multiple Output (MIMO) – multiple antennas on sending and receiving devices to reduce error and boost speed – this standard supports data rates up to 3.46Gbps. Some router vendors include technologies that support the 2.4GHz frequency via 802.11n, providing support for older client devices that may have 802.11b/g/n radios, but also providing additional bandwidth for improved data rates.

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- 802.11ax (Wi-Fi 6)
- Known as High Efficiency WLAN, 802.11ax aims to improve the performance in WLAN deployments in dense scenarios, such as sports stadiums and airports, while still operating in the 2.4GHz and 5GHz spectrum. The group is targeting at least a 4X improvement in throughput compared to 802.11n and 802.11ac., through more efficient spectrum utilization.

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Radio Frequency and the FCC

- Wireless primarily uses RF
  - Can interfere with critical applications
- Regulated strictly by the Federal Communications Commission (FCC)
  - Regulates what frequencies wireless communications can use, how much power antennas can emit, and other matters concerning the use of radio waves, infrared, and microwaves for communication
- When planning deployment, check with your local FCC office to learn about regulations or requirements you must meet

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Summary, Part 1

- Wireless transmissions use electromagnetic (EM) radiation, specifically radio frequency (RF) waves or infrared (IR) radiation, to communicate
- EM radiation travels in waves
- The RF spectrum is divided into bands based on frequency
- The speed and transmission range of a wireless network vary depending on the standard, equipment, environmental factors, number of users, location of clients, and purpose

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Summary, Part 2

- RF transmits a carrier signal
- RF data can be analog or digital
- Spread spectrum spreads a narrowband signal over a broader portion of the RF band
- Wireless network components include wireless NICs, access points, antennas, remote wireless bridges, and wireless gateways
- Antennas transmit and receive radio waves and can be omnidirectional, semidirectional, or highly directional

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Summary, Part 3

- A remote wireless bridge operates in four modes: access point, root, nonroot, and repeater
- IEEE 802.11 standards define three WLAN configurations: BSS, ESS, and IBSS
- IEEE 802.11 standards include: 802.11a, 802.11g, and 802.11n
- RF is subject to strict regulations by the FCC because of the potential for interference with critical communications, including radio, TV, military, and emergency services

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